

DS UA 9201 Problem Set 2

This homework must be turned in on Brightspace by **November 2nd 2022, 11:59pm (Paris time)**. It must be your own work, and your own work only—you must not copy anyone's work, or allow anyone to copy yours. This extends to writing code. You may consult with others, but when you write up, you must do so alone. Your homework submission must be written and submitted using Jupyter notebook. No handwritten solutions will be accepted. You should submit:

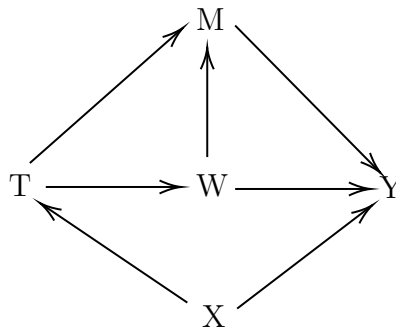
- A compiled PDF file named `yourNetID_solutions.pdf` containing your solutions to the problems.
- A `.ipynb` file containing the code and text used to produce your compiled pdf named `yourNetID_solutions.ipynb`. Note that math can be typeset in markdown cells (in the jupyter notebook) in the same way as LaTeX.

Please make sure your answers are clearly structured in the jupyter notebook file:

- Label each question (e.g. 1.1 for exercise 1, question 1).
 - Do not include written answers as code comments.
 - The code used to obtain the answer for each question part should accompany the written answer.
-

Exercise 1: Directed Acyclic Graph (25 points)

Consider the following Directed Acyclic Graph:



Question 1 Colliders (5 points)

Of the five variables in the graph, 2 are colliders with respect to a path and 3 are non-colliders. Which variables are colliders with respect to a path and which are non-colliders?

continued on page 2

Question 2 Identification (10 points)

Suppose that we wanted to estimate the effect of T on Y . Indicate if we should or should not condition on X , and explain why, and indicate if we should or should not condition on M and explain why.

Question 3 Paths (10 points)

Suppose that we wanted to estimate the effect of M on Y . List all the backdoor paths between M and Y , and indicate which variable we should condition on to block each path. There may be multiple valid options for each path.

Exercise 2: Social Pressure and Voter Turnout (again) (25 points)

Consider again the GOTV data from last problem set by Gerber, Green and Larimer (APSR, 2008). Although it is not specified in the paper, it is highly possible that the authors created subgroups based on the turnout history for 5 previous primary and general elections (number of times the individual voted), and number of registered voters in the household. In this problem, we will create subgroups based on the turnout history, and investigate the CATE (conditional average treatment effect) and the effect modifications in each subgroup. We denote the turnout history/number of times voted as a covariate X_i for individual i .

Question 1 Data Preparation (5 points)

Construct a new dataset for this problem using individual dataset from the last problem set.

1. Create a new column `num_voted` to represent the number of times the individual has voted in previous 5 elections by summing the variables `g2000`, `p2000`, `g2002`, `p2002` and `p2004` (exclude `g2004` because the experiment filtered out people who didn't vote in `g2004`), the resulting column should be an integer ranging from $[0,5]$
2. In the following problems, we are using the individual data with `num_voted` as different subgroups. To simplify the problem, we investigate only the "Neighbor" treatment effect. Construct a cleaner dataset with `id`, `hh_id`, `hh_size`, `num_voted`, `voted`, `treatment` as columns and filter out treatment groups besides `{Neighbor, Control}`.
3. Construct a household-level dataset by taking the means of `hh_size`, `num_voted`, and `voted` in each household (the other variables are all equal within the same household and can simply be left as they are). Round the mean of `num_voted` up to the nearest integer. Your resulting dataset should have one household

continued on page 3

per row, and hh_id, hh_size, num_voted, voted, and treatment as columns. The variable num_voted should have only values 0, 1, 2, 3, 4, 5.

4. Report number of households in each subgroup for both treatment and control, what do you observe?

Question 2 CATE for subgroups (10 points)

We define conditional average treatment effect as the ATE for different subgroups defined by the "num_voted" variable:

$$\tau(x) = E[Y_i(1) - Y_i(0) | X_i = x], \quad x \in \{0, 1, 2, 3, 4, 5\}$$

Since treatment was randomized at the household level, positivity and ignorability hold both unconditionally, and conditionally, within each subgroup. For each subgroup:

1. Estimate the CATE and report the variance of your estimates.
2. Construct an asymptotic 95% confidence interval around your estimates.
3. What conclusions can you draw from these statistics?

You can skip subgroups that either do not have members in them or do not have any treated/control members.

Question 3 Effect modification (10 points)

Suppose we want to estimate whether there is a difference in effects for two extreme groups, individuals who always vote ($X_i = 5$) and individuals who never vote ($X_i = 0$), we construct an estimator $\hat{\Delta}$ to estimate the difference. As we saw in class, we can estimate this difference as:

$$\hat{\Delta} = \hat{\tau}(0) - \hat{\tau}(5)$$

1. Calculate the variance of $\hat{\Delta}$ and construct a 95% confidence interval around it, can we say that there's significant difference in the treatment effect for people who always vote and people who never vote?
2. Combine your observations with conclusions from Question 2, comment about your findings.

Question 4 Sample sizes and significance effect (Bonus)

In the experiment, the authors claimed no significant differences between groups, one possible reason may be that the sample size for each subgroup is too small. This is a practical problem we may encounter in experimental designs when we are testing multiple hypothesis or we are having too many subgroups. Explain in your own words why having more hypothesis/subgroups would make significant effect harder to detect for each group, assuming the overall sample size is fixed.

continued on page 4

Exercise 3: Social Pressure and Voter Turnout (continued) (30 points)

In this question we will be using the same household-level dataset that you constructed in part a of Exercise 2.

Question 1 (5 points)

Compute the ATE of the "Neighbors" treatment using the standard difference-in-means estimator, i.e., $\hat{\tau} = \bar{Y}_t - \bar{Y}_c$. Provide standard errors and 95% confidence intervals for your estimates.

Question 2 (10 points)

Now compute the same ATE but with the stratification estimator that is defined as the weighted mean of the stratum CATEs that you computed in the previous problem:

$$\hat{\tau}_{\text{block}} = \sum_{x=0}^5 \hat{\tau}(x) \frac{N_x}{N}.$$

Compute variance and 95% confidence intervals for this estimator as well using the stratified variance estimator defined as:

$$\widehat{Var}(\tau_{\text{block}}) = \sum_{x=0}^5 \widehat{Var}(\tau(x)) \left(\frac{N_x}{N} \right)^2$$

Comment on the difference between the ATE estimates you obtained here and in part a and their variances. What is it due to?

Question 3 (15 points)

Now Divide the data set into 6 strata in such a way that each of the strata have same proportion of Treated and Control observations. You can do so by creating a new variable called "group" with values 0, 1, 2, 3, 4, 5 and randomly assigning each value to $N_t/6$ treated units and $N_c/6$ control units. You may exclude enough treated and control units from the data to make N_t and N_c divisible by 6.

Compute the ATE by applying the estimator $\hat{\tau}_{\text{block}}$ to these newly created strata. Provide variance estimates block and 95% confidence intervals for these ATE estimates as well using the stratified variance estimator. Is the variance of this estimator much different from that of $\hat{\tau}$ you computed in Question 1? Why do you think this is the case?

continued on page 5

Exercise 4: Estimand, estimator, bias, variance (30 points)

Question 1 Estimators for the population mean (10 points)

Let us consider identically distributed variables $\{X_1, X_2, \dots, X_n\}$ of mean μ and variance σ^2 .

Let us consider two estimators:

- the sample average $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$
 - the first observation (we disregard all the other observations) $x_1 = X_1$
1. A nice property of an estimator is unbiasedness. Show that both estimators are unbiased estimators of μ (i.e. their expectation is equal to μ).
 2. However, unbiasedness is not sufficient, and a criterion to distinguish a good estimator from a bad is its variance. Using basic properties of variance (variance of a sum of independent variables), establish the variance of each estimator in function of σ .
 3. Which estimator should we use in practice?

Question 2 Estimators for causal effect (20 points)

With the simple example of estimators for the mean, we have illustrated that we can derive statistical properties of estimators to help us select the best one. In the previous homework you have shown that the IPW estimator is unbiased. In this homework, you will be guided through deriving the variance of this estimator, again using simple properties of the variance.

We are going to derive the variance of the oracle IPW estimator, in which the propensity scores are known (as if we had an oracle to tell us), and not estimated. Indeed, propensity score estimations breaks the independence between observations as we use all samples to estimate them (through logistic regression), and this makes the computation very complicated.

We denote the oracle IPW estimator as

$$\hat{\tau}_{IPW}^* = \frac{1}{n} \sum_{i=1}^n \left(\frac{T_i Y_i}{e(X_i)} - \frac{(1 - T_i) Y_i}{1 - e(X_i)} \right)$$

We are going to show that

$$nVar[\hat{\tau}_{IPW}^*] = E \left[\frac{Y(1)^2}{e(X_i)} \right] + E \left[\frac{Y(0)^2}{1 - e(X_i)} \right] - \tau^2$$

We decompose the proof as following

1. Show that $Var[\hat{\tau}_{IPW}^*] = \frac{1}{n} \left(Var \left[\frac{TY(1)}{e(X_i)} \right] + Var \left[\frac{(1-T)Y(1)}{1-e(X_i)} \right] - 2Cov \left[\frac{TY(1)}{e(X_i)}, \frac{(1-T)Y(1)}{1-e(X_i)} \right] \right)$

continued on page 6

2. Show that $E \left[\left(\frac{TY(1)}{e(X_i)} \right)^2 \right] = E \left[\frac{(Y(1))^2}{e(X_i)} \right]$.
3. We admit similarly that $E \left[\left(\frac{(1-T)Y(0)}{1-e(X_i)} \right)^2 \right] = E \left[\frac{(Y(0))^2}{1-e(X_i)} \right]$. Using the general property of variance that $Var[W] = E[W^2] - E[W]^2$, show that $Var \left[\frac{TY(1)}{e(X_i)} \right] = E \left[\frac{(Y(1))^2}{e(X_i)} \right] - E[Y(1)]^2$ and that $Var \left[\frac{(1-T)Y(0)}{1-e(X_i)} \right] = E \left[\frac{(Y(0))^2}{1-e(X_i)} \right] - E[Y(0)]^2$.
4. We admit that $Cov \left[\frac{TY(1)}{e(X_i)}, \frac{(1-T)Y(0)}{1-e(X_i)} \right] = -E[Y(0)]E[Y(1)]$. Conclude the proof.

We could then use this result to compare the IPW estimator to other estimators for the ATE.